ACHREF DOULA

Postdoctoral Researcher — Reliable Machine Learning, Uncertainty & Human–AI Decision Making

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Research Profile

Machine-learning researcher working on uncertainty quantification, conformal prediction, reliable foundation models, and human–AI decision making. Research connects statistical guarantees under distribution and model shift with structured reasoning and uncertainty-aware interaction. PhD in Computer Science from TU Darmstadt (*summa cum laude*), with publications at ICML and accepted work at NeurIPS.

Academic Appointments

Postdoctoral Researcher and Project Lead **2025–Present**
Technical University of Darmstadt, Department of Computer Science *Darmstadt, Germany*

- Conduct and lead independent research on uncertainty quantification, conformal prediction, calibration under distribution shift, human–AI decision making, and reliable foundation models.
- Developed *Conformal Calibration Transfer* (ICML 2026), addressing calibration transfer across input spaces with finite-sample target-domain coverage guarantees and an observable diagnostic of transfer mismatch.
- Developed a framework for auditing conformal prediction for LLMs under inference-time model shift (NeurIPS 2026), studying how changes in inference configuration between calibration and deployment affect coverage guarantees.
- Developed a game-theoretic formulation of uncertainty in human–AI decision making that studies how uncertainty influences decisions to accept, verify, defer, or request additional information (ICML 2026).
- Lead the Ingenium-supported project *Uncertainty Quantification for Reliable Human–AI Interaction*, investigating uncertainty-aware policies for recommendation, verification, deferral, and information acquisition through controlled human-participant experiments.
- Supervise graduate researchers and coordinate collaborative experiments, research projects, and publications.

Affiliated Postdoctoral Researcher **Jul 2025–Present**
Tongji University, College of Architecture and Urban Planning *Shanghai, China*

- Conduct interdisciplinary research on LLM-based structured reasoning and decision processes for urban planning.
- Study SFT, DPO, and RLVR for reasoning quality, constraint adherence, and model evaluation.
- Investigate combinations of LLM reasoning with structured domain knowledge and neuro-symbolic constraints.

Doctoral Researcher and Project Lead **Jul 2020–2025**
Technical University of Darmstadt, Telecooperation Lab *Darmstadt, Germany*

- Developed uncertainty-aware methods for perception, prediction, planning, and human interaction in safety-critical autonomous systems.
- Introduced Knowledge-Refined Prediction Sets, combining conformal prediction with symbolic constraints for semantically consistent autonomous perception (CoRL 2024).
- Developed NeSyMoF, a neuro-symbolic approach to motion forecasting (IROS 2024).
- Developed SafePath, combining LLM-based planning with conformal uncertainty and human delegation.
- Led the BMBF-funded SemUrban project, including scientific direction, budget management, stakeholder coordination, and research transfer.

Education

PhD in Computer Science (Dr. rer. nat.), *summa cum laude* 2020–2025
Technical University of Darmstadt, Germany
Dissertation: *On Uncertainty-Aware Perception, Prediction, and Planning.*
Supervisor: Prof. Dr. Max Mühlhäuser.

M.Sc. in Electrical Engineering and Information Technology 2016–2020
Technical University of Darmstadt, Germany
Thesis: *Deep Active Learning for Semantic Segmentation.*

Dipl.-Ing. in Electrical Engineering 2010–2015
National Engineering School of Tunis (ENIT), Tunisia

Research Areas

Uncertainty and statistical learning: conformal prediction, uncertainty quantification, calibration, distribution and model shift, calibration transfer, finite-sample guarantees, epistemic uncertainty.

Rational and interactive AI: human–AI decision making, game-theoretic modeling, uncertainty-aware intervention, selective prediction, deferral, verification, information acquisition, repeated interaction.

Emerging research direction: possibilistic and imprecise representations of uncertainty for language models and their use in rational decision making.

Foundation models: LLM post-training, SFT, DPO, RLVR, structured reasoning, model evaluation, safety evaluation, agentic and retrieval-augmented systems.

Research Funding and Project Leadership

Ingenium Postdoctoral Support Package, TU Darmstadt 2026–2027
Uncertainty Quantification for Reliable Human–AI Interaction
EUR 3,000 — Applicant and Project Lead

- Independent research project on uncertainty-aware intervention policies and reliable human–AI decision making.
- Funding supports implementation and a controlled human-participant study examining decision quality, reliance, and human–AI team performance.

Software Campus Fellowship / BMBF Research Grant 2022–2025
SemUrban — Semantically Enriched Deep Learning for Autonomous Systems
EUR 100,000 — Project Lead

- Led the scientific agenda, technical milestones, budget, stakeholder coordination, reporting, and research transfer.

Selected Publications

1. **A. Doula**, M. Mühlhäuser. *Auditing Conformal Prediction for Language Models Under Inference-Time Model Shift.* Advances in Neural Information Processing Systems (NeurIPS), Evaluations & Datasets Track, 2026. **Accepted / forthcoming**. arXiv:2610.28937.
2. **A. Doula**. *Conformal Calibration Transfer.* Proceedings of the 43rd International Conference on Machine Learning (ICML), 2026.
3. **A. Doula**, O. Herzog, S. Z. Wu, M. Mühlhäuser. *Position: Uncertainty is a Strategic Signal in Human–AI Decision Making.* Proceedings of the 43rd International Conference on Machine Learning (ICML), 2026.
4. **A. Doula**, T. Güdelhöfer, M. Mühlhäuser, A. Sánchez Guinea. *Conformal Prediction for Semantically-Aware Autonomous Perception in Urban Environments.* Conference on Robot Learning (CoRL), 2024.
5. **A. Doula**, H. Yin, M. Mühlhäuser, A. Sánchez Guinea. *NeSyMoF: A Neuro-Symbolic Model for Motion Forecasting.* IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024.

Full publication list: Google Scholar.

Academic Service and Honors

- **Hessian AI Innovation Award**, 2026. Awarded to our project *4d.City: 4-Dimensional Counterfactual Urban World Models* on 29 October 2026 (team award).
- **NeurIPS Top Reviewer Award**, 2026.
- **ICML Gold Reviewer Award**, 2026.
- Reviewer for TMLR, ICML, NeurIPS, KDD, ICCV, NAACL, ICRA, CHI, and IEEE Robotics and Automation Letters.
- Supervised **10+ Bachelor's and Master's theses** across uncertainty quantification, conformal prediction, neuro-symbolic learning, autonomous systems, LLM-based systems, and human–AI interaction.
- Teaching contributions to *Distributed Algorithms* and *Ubiquitous Computing / Machine Learning.*

Earlier Industry Research Experience

Working Student, Deep Learning **May 2019–Jun 2020**
Merck KGaA, Big Data and Advanced Analytics *Darmstadt, Germany*

Developed and evaluated deep-learning approaches for pharmaceutical knowledge graphs, drug-discovery applications, and molecular-property prediction.

Working Student, Deep Learning **Oct 2018–Apr 2019**
IFCC GmbH *Dieburg, Germany*

Developed deep-learning models for industrial product classification and anomaly detection.

Research Methods and Technical Skills

Methods: conformal prediction, uncertainty quantification, calibration, statistical learning under distribution shift, game-theoretic modeling, human–AI experimentation, neuro-symbolic learning, LLM post-training and evaluation.

Machine learning and foundation models: Python, PyTorch, Hugging Face Transformers, SFT, DPO, RLVR, PEFT/QLoRA, RAG, embeddings, tool calling, reasoning and safety evaluation.

Languages: Arabic (native), French (C2), English (C1), German (C1), Italian (A2).